

Goodness-of-fit Diagnostics for State Space Models in Neuroscience

Sirui Zeng¹, Uri T. Eden¹, and Eric L. Denovellis^{2,3,4}

¹Department of Mathematics & Statistics, Boston University, Boston, Massachusetts

²Howard Hughes Medical Institute, University of California, San Francisco, San Francisco, California

³Departments of Physiology and Psychiatry, University of California, San Francisco, San Francisco, California

⁴Kavli Institute for Fundamental Neuroscience, University of California, San Francisco, San Francisco, California

Correspondence: `eric.denovellis@ucsf.edu`

November 11, 2025

Abstract

State space models are widely used in neuroscience to relate neural activity to latent dynamic brain states. This paper introduces diagnostics (KL divergence and HPD overlap) to assess model goodness-of-fit by examining consistency between posterior distributions and component likelihood distributions.

Keywords: state space models, goodness-of-fit, model diagnostics, Bayesian inference, neuroscience, neural decoding

1 Introduction

2 Methods

2.1 State Space Model Framework

2.2 Diagnostic Metrics

3 Results

3.1 Simulated Data

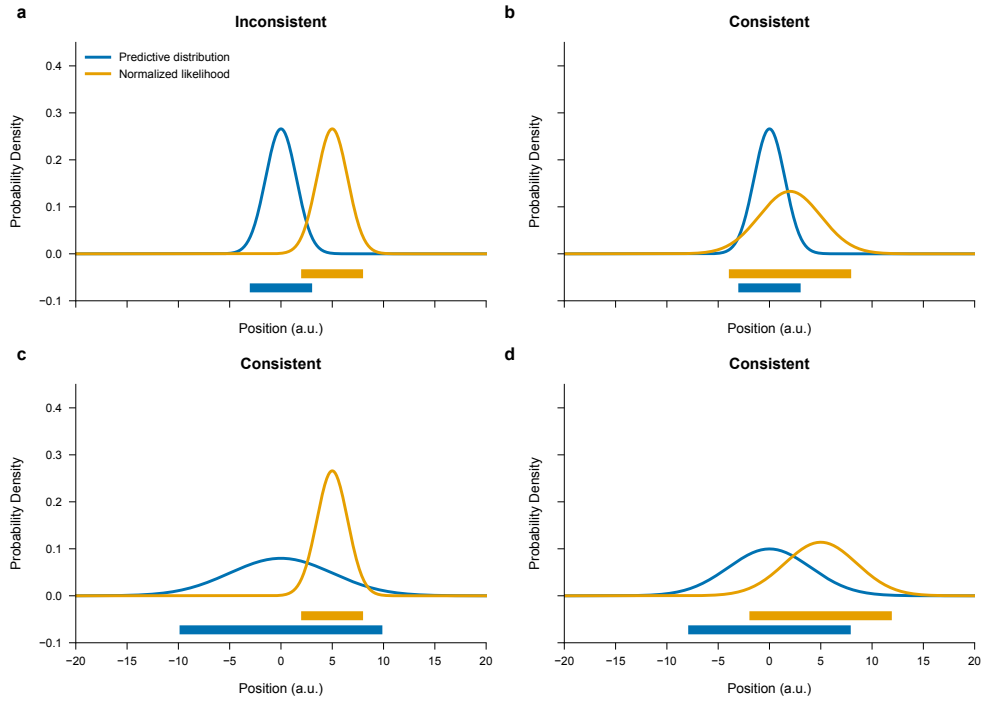


Figure 1: **Distribution consistency examples.** Four scenarios showing predictive distribution and normalized likelihood with varying degrees of consistency.

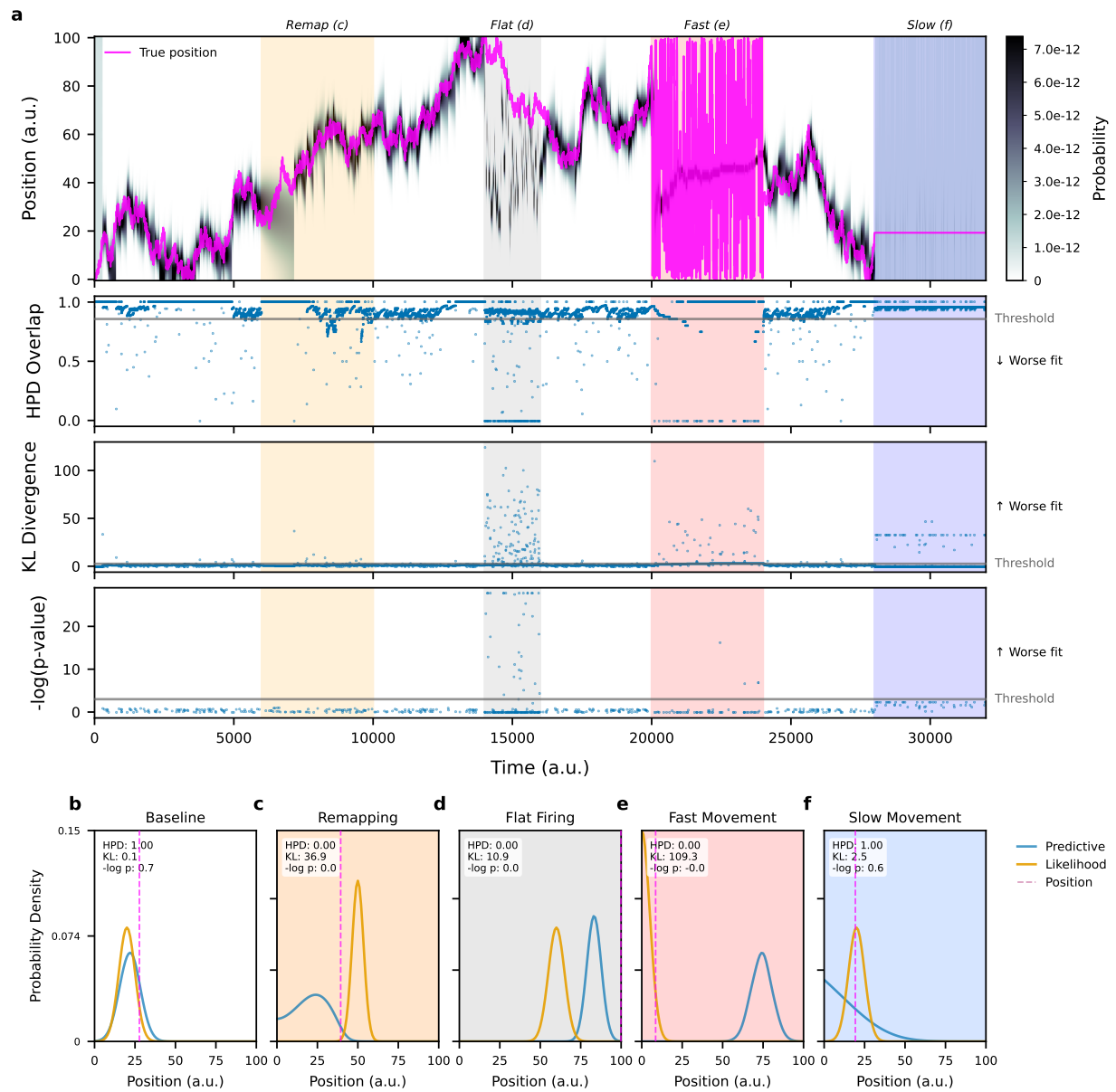


Figure 2: **Diagnostic demonstration on simulated data.** Time-resolved diagnostics showing periods of good and poor model fit.

14 4 Discussion

15 5 Conclusion

16 Code and Data Availability

17 All code to reproduce figures and analyses is available at [https://github.com/Eden-Kramer-Lab/](https://github.com/Eden-Kramer-Lab/statespacecheck-paper)
18 [statespacecheck-paper](https://github.com/Eden-Kramer-Lab/statespacecheck-paper). The `statespacecheck` Python package is available at [https://github.](https://github.com/Eden-Kramer-Lab/statespacecheck)
19 [com/Eden-Kramer-Lab/statespacecheck](https://github.com/Eden-Kramer-Lab/statespacecheck).

20 Author Contributions

21 SZ and ELD designed the study. SZ developed the diagnostic methods. ELD implemented the
22 software package. SZ and ELD performed the analyses. SZ, UTE, and ELD wrote the manuscript.

23 Competing Interests

24 The authors declare no competing interests.

25 Acknowledgments

26 References